

X-Ray Images Classification into Normal, Pneumonia and Tuberculosis

Final Project Report – Applied AI in Biomedicine

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1. INTRODUCTION

Pneumonia and Tuberculosis (TB) are two of the most prevalent and deadly pulmonary diseases worldwide, causing significant morbidity and mortality in both developing and developed countries.

Pneumonia is a serious lung infection caused by a variety of pathogens, including bacteria and viruses. It inflames the alveoli and fills them up with fluid or pus possibly leading to a decrease of inhaled oxygen concentration in the patient's bloodstream. TB instead is caused by a bacterium, i.e. *Mycobacterium tuberculosis*, and is primarily spread through the air through the inhalation of droplets from an infected person. Even if diagnosis and treatment of pneumonia and tuberculosis have improved significantly in the last decades, they still remain significant public health concerns. One of the main tools commonly employed for pneumonia and TB diagnosis is chest X-ray (CXR). However, CXR lung diseases patterns present difficulties in differentiation and may result in inter-reader variability across radiologists [1], while a precise and early diagnosis of such diseases is fundamental for the right treatment and the healing of the patient. Classification and detection of anomalies in XR images using machine learning and deep learning has shown promising performances and has become a very popular area of research. The main goal is to design reliable AI tools that could both help the clinicians perform diagnosis and speed up the screening phases. In this report, we present and analyse some deep learning approaches that could be applied in classification of CRX images into Normal, Pneumonia and TB.

2. MATERIALS AND METHODS

2.1 DATASET ANALYSIS

The dataset used for this experiment is composed by 15470 posteroanterior chests X-Ray images divided into three

classes. In particular, the set consists of 9354 *Normal* images (60.47%), 4250 *Pneumonia* images (27.47%) and 1866 *Tuberculosis* images (12.06%). A representative sample for each class is reported in Figure 1. The collected scans were not uniform in size. Furthermore, since patient data, like gender or age, were not available, these features were not considered in this study.

The data collected are related to 12086 patients, value that does not correspond to the total number of images contained in the dataset. In fact, it was noticed that some patients (approximately the 30% for all the classes) were related to two images representing the same posteroanterior view (double images), while others to only one (single images). The detailed distribution of images and patients is illustrated in Figure 2. With a further analysis, it was also assessed that the same patient did not appear in more than one class. Hereafter, some properties and irregularities encountered both among single and double images are reported. Firstly, the dataset analysis showed that not all the images were in *standard radiological format*, with the bones appearing white and the background black: indeed a further assessment revealed that 2561 scans over 15470 (16,6 %) were provided in the “negative” format, with black bones and white background. This unbalance between the two formats was handled in the preprocessing phase in order to avoid a possible bias in the model. Furthermore, 3056 images were found to be *noisy or corrupted*. Specifically, three types of noise/irregularity were identified: salt and pepper noise,

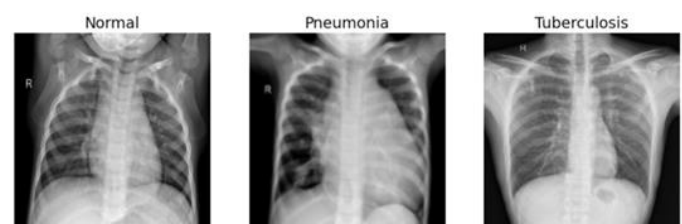


Fig.1 X-ray images from each class

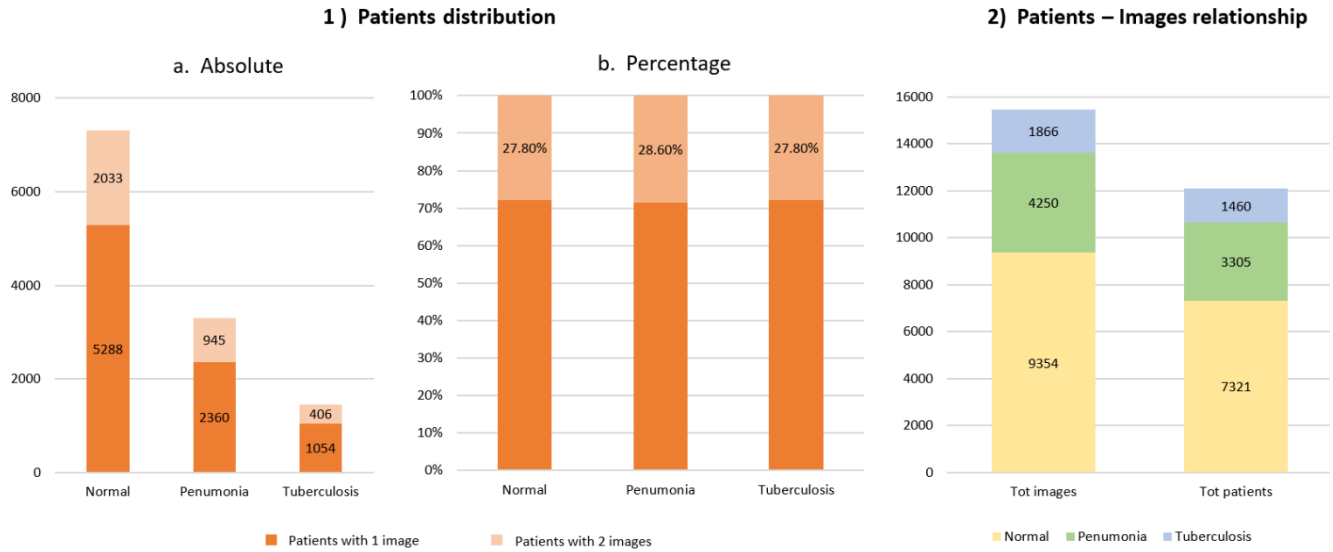


Fig. 2 1) Patients distribution, with respect to their related number of images among the three classes, in absolute value (1.a) and percentage (1.b) 2) Comparison between the distribution of the total images in the initial dataset and the total number of patients for each class: 12086 patients correspond to the original 15470 scans.

uniform-like noise and a particular corruption of the image characterized by the presence of black spots (see Figure 3). Lastly, another feature, which was not deemed relevant at the beginning but that became so after analysing models performances, was the presence of letters in the picture which were either indicative of the orientation of the chest or related to other technical information/patient data. To avoid that classification was biased by the presence of these letters, we removed them in the preprocessing phase.

2.2 IMPLEMENTATION

The principal steps of this study can be reassumed in 1) Data Preprocessing, 2) Learning model definition and training, 3) Explainable AI implementation. Concerning the data preprocessing, three different approaches were investigated and compared after the training phase. All the pipeline was constructed using Keras library and Python programming language and it was run on Google Colaboratory platform.

2.3 DATASET PREPROCESSING

To provide the learning model with cleaned and enhanced pictures, different techniques have been employed. This

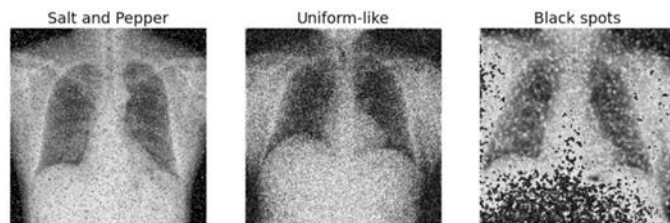


Fig. 3 Different types of noise affecting images

section examines the specifics of each image preprocessing procedure applied to the dataset.

2.3.1 IMAGE RESIZING AND TEXT REMOVAL

Initially, since the images in the dataset were not uniform in size, they were all resized into $256 \times 256 \times 3$. Moreover, since we noticed that letters, often included in the scans, were biasing the decisional process of the classifier, i.e. letters were considered as main features, they have been removed exploiting the text recognition tool provided by Keras named *Keras-ocr*. After recognizing the letters, they are surrounded by a box; then the content of the box is deleted and subsequently replaced by a pattern consistent with the adjacent regions of the image through the function *inpaint* provided by the library OpenCV [2].

2.3.2 IMAGE RADIOLOGICAL FORMAT

To assess the type of radiological format of the images, standard or negative, a background analysis was performed. First, six different Regions of Interest (ROIs) were defined along the edges of each picture. The upper left and lower right ROIs were not placed at the corresponding picture corner since the dataset inspection revealed the presence of irregularities in those areas. Specifically, they were due to the slight blurring that could result from the text removal process applied in those extremal regions of the image where letters mainly appeared. To continue, the mean pixel value for each ROI was computed (pixel intensity range 0 – 255). The image format was considered standard if at least 2 ROIs presented a mean lower than 85. At this point, to have all the pictures in the standard format (see 2.4.2), the negative ones,

i.e. images not verifying the imposed condition, were converted applying, for each pixel of the image, the transformation $(255 - \text{pixel value})$. A sample in the standard radiological format with the associated ROIs has been reported in Figure 4.

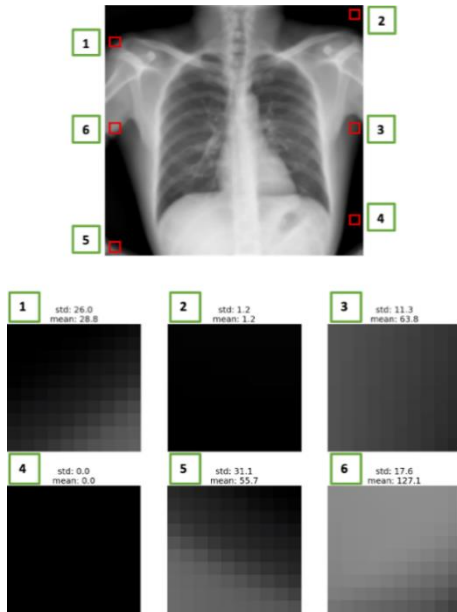


Fig.4 Image in the standard radiological format and its ROIs (in red) with the associated mean and standard deviation

2.3.3 IMAGE DENOISING

Image denoising consists in reducing noise, thus restoring the true image without losing its features. In a first phase we performed a histogram analysis to find possible patterns that could be useful in distinguishing between the three types of noise. Since this analysis was not effective, we considered as discriminative feature to individuate the presence of noise the standard deviation of each of the six ROIs, used also to define the radiological format. In this case again the results didn't show a specific trend for each typology, but we were able to define the following approach for the final evaluation. If the standard deviation resulted to be higher than 20 for all the ROIs, the image was assessed to be noisy, regardless the class of noise. After the identification of the noisy scans, different low-pass filter kernels were investigated and implemented to try to filter out the noise: *Mean*, *Median*, *Gaussian*, *Bilateral filter* [3]. The median filter is the one that produced better results for at least one type of noise. The other filters, instead, resulted to distort much more the images without effectively removing any noise. Even if the procedure described at the beginning of the paragraph was not able to recognize all the noisy images, we decided to apply a 5×5 *median filter* just to the pictures detected by it. In fact, since the spatial filters eliminate high frequency noise at the expense of blurring fine details and sharp edges, in this

way we were able to preserve images that were not affected by noise.

2.3.4 IMAGE ENHANCEMENT

Improving the quality of low contrast images is a key aspect in image processing, and image enhancement is a crucial technique used to achieve this goal. Specifically, among the existing methodologies, *Histogram Equalization* (HE) has been employed [4]. The central concept behind this method consists in redistributing pixels' intensities to achieve the most uniform distribution possible. In order to implement this procedure and obtain the results illustrated in Figure 5, the *equalizeHist* function of OpenCV library was used.

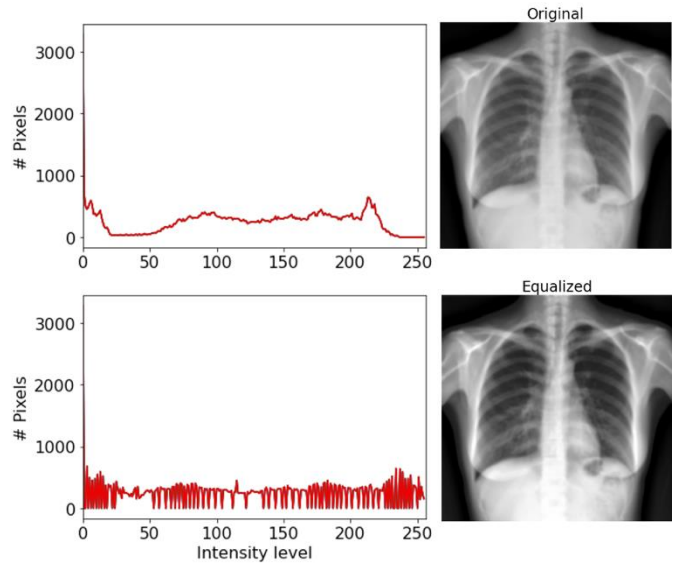


Fig.5 Example of image equalization with the associated histograms.

2.3.5 DATA AUGMENTATION

To expand the dataset and increase the robustness of the model, we applied augmentation techniques consisting in some basic image manipulation. Specifically, four augmentation strategies were investigated in this study: *Horizontal flip*, *Zoom*, *Vertical* and *Horizontal shift*. All the relative parameters can be found in Table 1.

	Zoom	Vertical shift	Horizontal shift
Range	0.1	0.05	0.05

Table 1 Augmentation parameters

2.4 APPROACHES DEFINITION

Once defined the most powerful preprocessing techniques, we decided to implement and compare three different approaches that resulted from considering different data

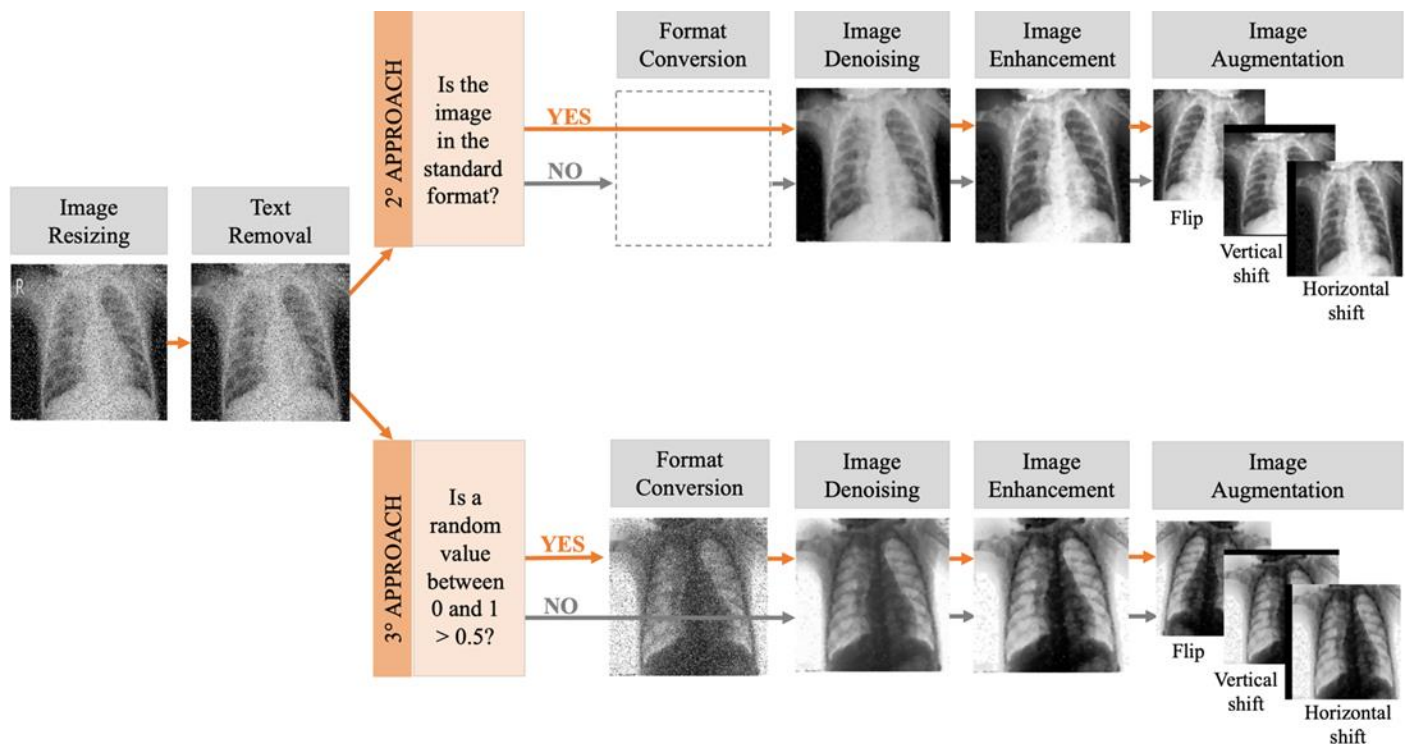


Fig.6 Summary diagram illustrating all the preprocessing steps adopted for second and third approach. The sample image under investigation is in the standard format (black background), noisy and presents a letter to be removed. According to the approach considered the image will be processed in a different way. Results illustrated in the pictures are referred to the steps indicated by the orange arrows.

taken from the original dataset and from combining part of the preprocessing techniques previously described.

2.4.1 FIRST APPROACH: BASELINE

In this approach the entire dataset was considered (15470 images) and just the median filter was applied. Since this last preprocessing tool was employed also in the other two approaches, this first procedure can be referred as the *baseline* methodology.

2.4.2 SECOND APPROACH

For the second approach not all the images in the dataset were considered. As stated before, there were patients associated to couple of images therefore, just one picture for patient was selected to make the model assessment more reliable: in fact, with the number of scans unbalanced between patients, the results could be biased. In particular, two types of couples have been identified: the first one containing a picture in standard format and a noisy one, the second type containing pictures in the two different formats. Among couples of the first type, we kept the noisy image to make the model more robust with respect to them, and among those of the second type we kept the standard one, resulting in an initial set of 12086 images. Regarding the preprocessing, we performed the following steps: resizing, text removal, conversion of all the images in the negative

format into the standard format, denoising with the median filter of the noisy scans, histogram equalization. Data augmentation, described in the paragraph 2.3.5, has been applied to the training set as last step.

2.4.3 THIRD APPROACH

The implementation of the last method started from the same initial dataset of the second approach, considering the same picture for each patient (dataset of 12086 images). In this case we operated such that images were not all in the same format but were equally distributed between the two categories. However, since the distribution was heavily unbalanced with many more images in the standard format with respect to those in the negative format, we transformed each image into its opposite (e.g. from standard to negative format and vice versa) with a probability of 50%. In this way, the dataset has been forced, in a probabilistic way, to be composed by images as much as possible equally distributed between the two formats. All the other preprocessing steps were applied in the same way of the second approach.

2.5 MODELS

After the preprocessing we moved to the following phase consisting in the model definition and training. With respect to the model structure, we tried several options mainly based on Convolutional Neural Networks.

2.5.1 EFFICIENTNETB3

The model characterized by the best performance exploited EfficientNetB3 as base model which turned out to be a good trade-off between architecture complexity and performance quality. It is a deep learning model provided by Keras alongside pre-trained weights obtained with the ImageNet dataset that we decided to maintain instead of starting with a complete random initialization.

We set the parameter *include_top* = *False* to customize the Fully Connected layers to be put on top of the CNN. The Fully Connected part started with a Dropout layer with rate = 0.4 (fraction of the input units to drop), which turns off activation values in the maps of the final convolutional output, modifying the averages calculated by the following layer, i.e. Global Average Pooling (GAP) layer, thus improving generalization capability. We went on adding other layers i.e. Dropout layer (rate = 0,4) – Dense layer (256 neurons) – Dropout layer (0,4) – Output Dense layer (3 neurons encoding the 3 classes). We noticed that dropping the same portion of units in all Dropout layers was preferable with respect to gradually decrease the rate, so we went ahead maintaining the same rate. ReLu and SoftMax were the activation functions used respectively in the hidden dense layer and in the output layer. Moreover, in the two dense layers, we used kernel initializers to set the initial random weights according to a HeUniform distribution in the first one and to a GlorotUniform distribution in the second one (output layer). In the latter, we also included a kernel regularizer that applies a L2 regularization penalty to prevent overfitting. Besides this network we also tried other pre-trained models even more complex in terms of architecture and consequently of number of parameters. One of them was ConvNeXtLarge which has way more parameters with respect to EfficientNetB3 (197.7 millions of parameters versus 12.3 millions of EfficientNetB3).

2.5.2 DEEP LEARNING MODEL FROM LITERATURE

Another option consisted in exploiting a network found in literature [5] used for a similar classification task, i.e. classification of chest X-Ray images into COVID-19, Pneumonia and Tuberculosis. The convolutional part of the model is made by a block, repeated four times, which is constituted by the layers Conv2D, MaxPooling and Dropout. On top of this, there is the Fully Connected part which starts with a Flattening layer, goes on with a Dense layer followed by a Dropout layer, to improve generalization, and ends with a final Dense layer, including 3 output neurons, from which class probabilities are derived.

2.5.3 MACHINE LEARNING CLASSIFIER

The last attempt consisted in exploiting a pre-trained network such as EfficientNetB3 to perform the initial feature

extraction process, and then training a Machine Learning (ML) classifier to make predictions.

Specifically, after deriving the feature vector for each sample of the training set using the convolutional portion of the CNN, the ML classifier (we tried with Random Forest and AdaBoost) has been fitted on these data and then has been used to make predictions on the test set whose features had been previously extracted as well. We couldn't try more powerful ML models, such as SVM, due to Colab RAM limited capacity.

2.6 TRAINING IMPLEMENTATION

First, we divided the dataset performing a stratified random split into train, validation and test sets with the percentages of 80%, 20% and 10% respectively. Regarding the first approach, we avoided that pictures belonging to the same patient appeared in different sets; concerning the last two options, we made sure that patients present in each set were the same for both the approaches, in order to make a comparison as coherent as possible.

In order to train our best model based on EfficientNetB3, we decided to apply sequentially both Transfer Learning and Fine Tuning. ImageDataGenerator class has been exploited to import images from the pre-made folders of training, validation and test set and to simultaneously perform image augmentation on the training set. For the Transfer Learning (TF) part the listed steps have been implemented:

1. Instantiate a base model (EfficientNetB3) and load pre-trained weights into it (ImageNet weights)
2. Freeze all layers in the base model by setting the parameter *trainable* = *False*
3. Put the FC part on top of the output of the base model, making sure to pass the parameter *training* = *False* when calling the base model, so that it runs in inference mode and *batchnorm* statistics don't get updated even after we unfreeze the base model for fine-tuning [6]
4. Train the top layer with our dataset; since it was a multiclass classification problem we used Adam optimizer (weights are adjusted based on the moving average of gradients calculated in current and previous epochs, [7]) with a fixed learning rate of 1e-3 and Categorical Crossentropy as loss function.

Even at this early stage we exploited Early Stopping to reduce overfitting without compromising model accuracy. Moreover, to make the model pay particular attention to under-represented examples, we introduced class weights so that the prediction error would be larger when an instance belonging to an under-represented class is not properly

classified. Specifically, we defined the weight for the i^{th} class according to the following formula:

$$\text{class weight}_i = \frac{n^{\circ} \text{ samples tot}}{n^{\circ} \text{ samples}_i}$$

where the numerator indicates the number of samples composing the training set and the denominator the amount of samples belonging to the i^{th} class; thus the smallest the amount of data for a certain class the higher the associated weight.

Then we moved to the Fine-Tuning (FT) part; since EfficientNetB3 was composed by several concatenated blocks, we tried to unfreeze gradually an increasing number of blocks assessing each time the achievable accuracy. At the end we opted for unfreezing from layer 118 (block 4) of the network. In addition to the previous settings, during this second phase we imposed a smaller initial learning rate (i.e. $1e-5$) which was no longer fixed but was reduced according to the ReduceOnPlateau monitoring option. Specifically, the learning rate was halved if after 5 epochs the validation loss did not decrease. Early Stopping was set with a patience equal to 20 such that the training would be arrested only after the appropriate stabilization of the loss function.

2.7 TEST TIME AUGMENTATION

At inference time we tried to further improve our results implementing Test Time Augmentation (TTA); similar to what Data Augmentation is doing to the training set, the purpose of TTA is to perform random modifications to the test images. Thus, instead of showing the regular image only once to the trained model, we show it the augmented images several times with the same transformations performed during training [8]. We then averaged the predictions of each corresponding image and took that as our final guess.

2.8 XAI METHODS

Once implemented our model, we recurred to Explainable Artificial Intelligence (XAI) to perform a further evaluation. In particular, XAI is a set of processes and methods that allows human users to comprehend and trust the results and output created by machine learning algorithms [9]. Among them we opted for three different techniques i.e. GradCam, LIME and Occlusion method.

Grad-CAM, presented by Selvaraju and coauthors [10], uses the gradient of the classification score with respect to the convolutional features determined by the network in order to understand which parts of the image are most important for classification.

The explainable AI method LIME (Local Interpretable Model-agnostic Explanations) manipulates the input data and creates a series of artificial data containing only a part of

the original attributes, thus in case of images it deletes each time a different group of pixels and performs prediction to see how relevant that area was for classification purposes [11]. The last technique, i.e. Occlusion method, was theoretically comparable to the previous one but in this case the “occluded” pixels assume a geometrical shape, in our case a square of 64x64 pixels, that is moved time by time along all the image with a stride of 16 pixels.

All the methods have been applied to a random set of images to understand if the algorithm exploited the most representative regions of the image which are different according to the considered condition, i.e. healthy, pneumonia and tuberculosis.

3. RESULTS AND DISCUSSION

3.1 MODELS

In this study, we evaluated the performance of machine learning models and deep learning models. The models tested included EfficientNetB3 as feature extractor with Random Forest and AdaBoost, the CNN developed by Bhandari et al [5], EfficientNetB3 and other more complex deep NN such as ConvNeXtLarge with transfer learning and fine tuning. Our results showed that the EfficientNetB3 model with TF and FT performed the best in all the three approaches, with accuracies always above 0.93 and TB F1 score above 0.82 in both validation set and test set. The most complex deep NN presented a higher overfitting of train set and slightly worse performances in terms of validation and test sets accuracies. The ML approach with Random Forest and AdaBoost was discarded as the F1 scores did not reach the threshold of 0.75 required for TB F1 score. Finally, the performances of the CNN from literature were slightly poorer with respect to those of the network based on EfficientNetB3 and moreover, at early stages of this study, before applying the text removal algorithm, the model tended to be more biased by the letters in the images. In light of the obtained results, we decided to pursue our study with the EfficientNetB3 model. The final relative FC structure was the one described in 2.5.1 and the FT process was performed unfreezing from layer 118.

3.2 THREE APPROACHES COMPARISON

The classification accuracies obtained with EfficientNetB3 model for each of the three approaches described in 2.4 are shown in Table 2.

Approach	Training set	Validation set	Test set
1	0.9946	0,9460	0,9452
2	0.9881	0,9524	0,9513
3	0.9681	0,9392	0,9389

Table 2 Accuracies

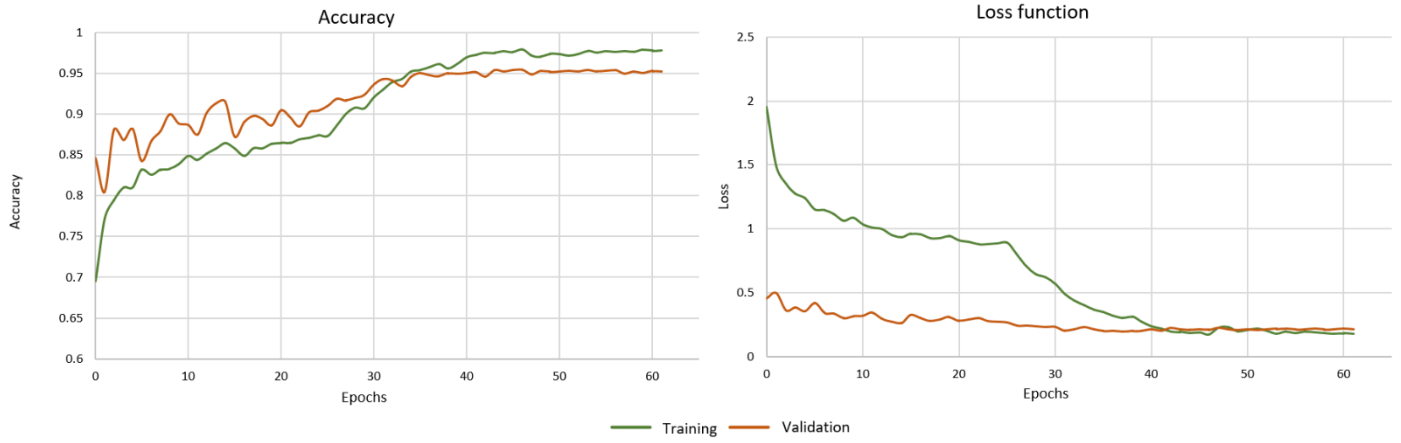


Fig.7 Trend of accuracy (Left) and loss function (Right) for training and validation during Transfer Learning (first 40 epochs) and Fine Tuning

Although its good performance, the first approach was discarded since we deemed that considering just one picture for patient was more reliable in the evaluation phase. Among the remaining two, the best approach turned out to be the second one: the network might have learnt better the features of images when they were all in standard format with respect to the case in which they were both in standard and negative format. The graph of the accuracy and the loss function for training and validation are reported in Figure 7, while the F1 scores relative of this approach are displayed in Table 3. In general, the results obtained in the second approach present a good performance and a good coherence between validation and test sets. The increase of TB F1 score from validation to test set could be due to the random splitting in the three sets: TB test set might be more easily classifiable. Furthermore, it can be noticed that the class with greater classification error is TB: this could be due both to its underrepresentation in the whole dataset and to some intrinsic characteristics of TB CRX that might be hard to learn for the network.

	Normal		Pneumonia		TB	
	-	TTA	-	TTA	-	TTA
Validation	0.96	0.96	0.98	0.98	0.83	0.84
Test	0.96	0.97	0.96	0.96	0.86	0.87

Table 3 F1 scores in approach 2 before and after TTA

3.3 TTA RESULTS

At inference time we tried to further improve our results implementing TTA on our best model of the second approach. The validation and test accuracies presented a small increase, reaching values of 0.9549 and 0.9553 respectively. The F1 scores are showed in Table 3, in

particular TB showed an increase of 0.01 points in both sets. This behavior. i.e. overall accuracy remains almost the same while TB F1 increases more, can be explained as follows: the general performance is more influenced by Normal and Pneumonia classes as they are more numerous, and these presented already very good F1 reaching a plateau, while TB had more margin of improvement but not much influence on the overall accuracy.

3.4 MODEL EXPLANATION WITH XAI

We adopted three different techniques of XAI for model interpretation: GradCam, LIME and Occlusion method. In Figure 8 the three methods are displayed for each class. Analyzing the results for some samples, we noticed that the regions more targeted by these algorithms were located in the middle-top area of the lungs for TB and in the middle-low for Pneumonia, while for Normal the regions were spread in all the lungs area. In general, among the three approaches, GradCam resulted to be the more interpretable model, showing for the majority of the images reasonable parts that were considered to make a prediction, helping to understand the decision-making process of the model. On the other hand, LIME and Occlusion methods were more unpredictable.

4. CONCLUSION AND FUTURE DEVELOPMENTS

In this study we proposed a CNN model for CXR image classification into normal, pneumonia, and tuberculosis, and included a XAI framework for explanation generation. Results of this investigation, based on transfer learning and fine tuning techniques, show that using image preprocessing, enhancing and balancing techniques can potentially increase the model's performance. In fact, the following pipeline allowed our proposed model to reach 95% of accuracy at inference time: balancing of dataset with respect to number of images for patient, text removal, standard format

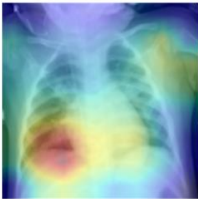
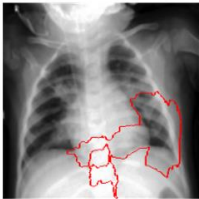
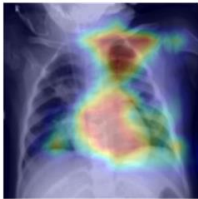
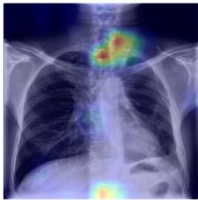
	Image	Gradcam	LIME	Occlusion method
Normal				
Pneumonia				
Tuberculosis				

Fig.8 Comparison of XAI methods results for one image from each class

conversion, image denoising (median filter), image enhancement (HE), data augmentation (zoom, flip, shift) and finally TTA. Moreover, the lower performing class (Tuberculosis) was still able to achieve an f1 score of 83%. The experiment was executed considering always the same set of images from the dataset, both at training and inference phase. For this reason, k-fold validation could be performed as possible improvement: this method, that was not implemented due to computational reasons, could improve the performance increasing the generalization capabilities of the model. Concerning the preprocessing, segmentation could be another possible approach: the lung's area could be detected and selected, and the following steps of the pipeline could be implemented considering just this region. In conclusion, our design has the ability to automatically identify distinctive features in X-rays images without the need for manual feature selection, and it has demonstrated outstanding results.

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